



Title: AI-BASED NETWORK ROUTING OPTIMISATION

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Introduction

[The rapid growth of IoT devices has increased network traffic and created challenges such as congestion, delay, and inefficient routing. Traditional routing protocols use fixed shortest-path decisions and cannot adapt to changing network conditions. This project proposes an AI-driven routing system that combines Software Defined Networking (SDN) with a Dueling LSTM-DQN agent. The system continuously monitors network conditions and dynamically selects the best routing path for different traffic types. This improves latency, reliability, throughput, and overall network performance in IoT

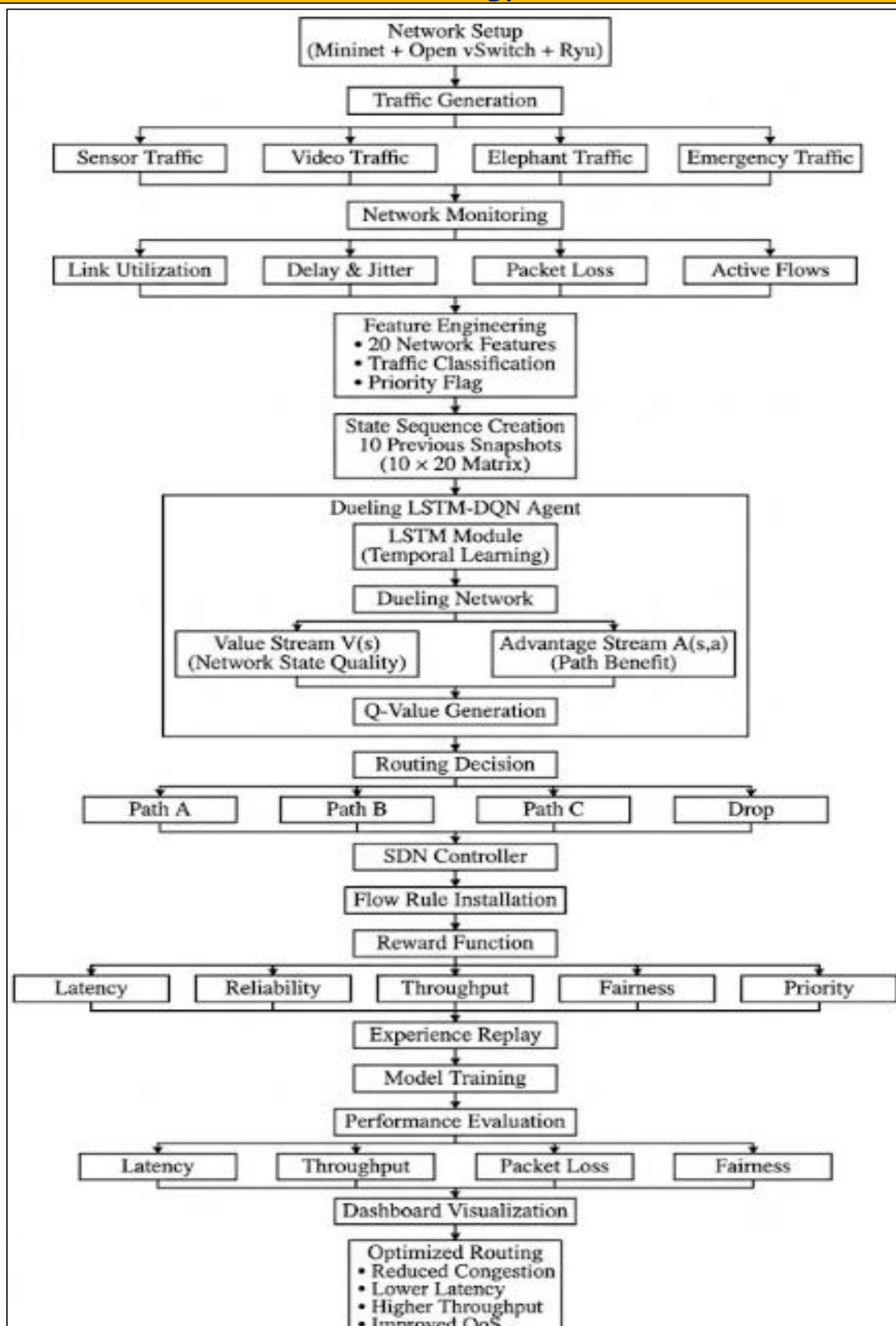
Problem Definition

IoT networks carry diverse traffic such as sensor data, video streams, bulk transfers, and emergency alerts, each having different network requirements. Traditional routing protocols rely on fixed shortest-path decisions and cannot adapt to changing network conditions, leading to congestion, increased delay, packet loss, and inefficient resource utilization. An intelligent and adaptive routing mechanism is required to optimize path selection in real time while prioritizing critical traffic.

Objectives

- Develop an AI-based SDN routing system.
- Monitor real-time network conditions.
- Implement Dueling LSTM-DQN for routing decisions.
- Reduce latency and packet loss.
- Improve throughput and bandwidth utilization.
- Prioritize critical and emergency traffic.
- Optimize routing under congestion.

Methodology



Tools used

- Python - Implementation of AI algorithms and network analysis.
- PyTorch - Building and training the Dueling LSTM-DQN model .
- Mininet - Creating and simulating the SDN network topology .
- Open vSwitch (OvS) - Software-defined switch for packet forwarding .
- Ryu Controller - Centralized SDN controller for network management .
- OpenFlow Protocol - Communication between controller and switches .
- Flask REST API - Interface between Ryu Controller and AI agent .
- Linux(Manjaro/Ubuntu) - Operating system for deployment and testing .
- Web socket / SEE - Real-time network statistics update.

Results and Discussions

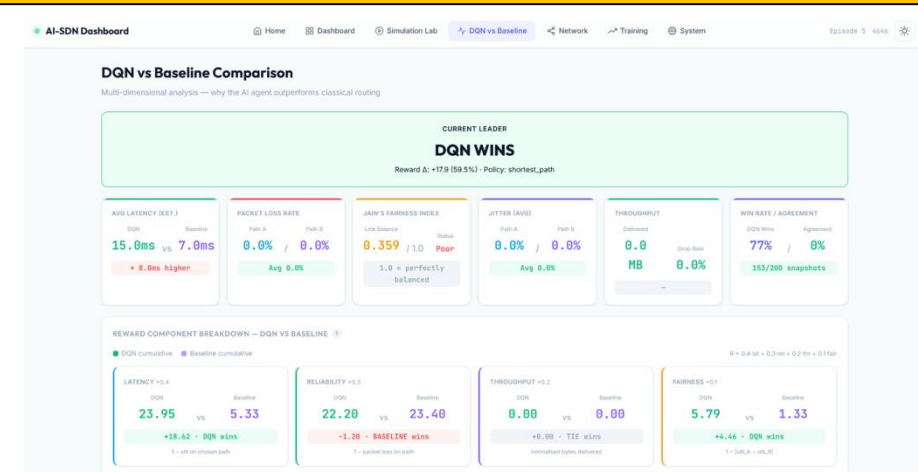
The proposed Dueling LSTM-DQN based SDN routing framework successfully learned to adapt routing decisions according to real-time network conditions. Experimental results showed improvements in latency, throughput, congestion management, and overall network efficiency compared to traditional routing approaches. The LSTM component effectively captured traffic trends, while the Dueling DQN architecture enabled better path selection under dynamic traffic scenarios.

- Reduced average network latency through intelligent path selection.
- Improved throughput by utilizing less congested network routes.
- Lower packet loss during high traffic and congestion conditions.
- Successfully prioritized emergency and delay-sensitive traffic.
- Achieved balanced resource utilization across multiple network paths.
- Outperformed baseline routing policies in cumulative reward and routing efficiency.
- Demonstrated stable learning behavior using Experience Replay and Double DQN techniques.

Conclusions

This project presented an intelligent SDN-based routing framework that integrates a Dueling LSTM-DQN model for dynamic network path selection. By leveraging real-time network statistics and historical traffic patterns, the system was able to make adaptive routing decisions that improved overall network performance. The proposed approach effectively reduced latency and packet loss, increased throughput, and prioritized critical traffic under varying network conditions. Experimental results demonstrated that the AI-driven routing strategy outperformed traditional routing methods and provided better Quality of Service (QoS) in IoT environments. The project highlights the potential of combining Software Defined Networking and Reinforcement Learning to build efficient, scalable, and intelligent next-generation networks.

Outcome of the work



References

